# Anmol Srivastava

**Backend Engineer · Python · FastAPI · RAG · LLM Integration**

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## Education

**United University**: Prayagraj, Uttar Pradesh 2023 – 2026

Bachelor of Computer Applications (in collaboration with IBM), Specialization in Artificial Intelligence

## Experience

**BNCW**: New Delhi, India Sep 2026 – Present

*Backend Engineer (Full-time)*

- Develop backend services for a production AI stock-research SaaS platform for a US-based client, spanning 3 FastAPI services (core API, RAG chatbot on PostgreSQL + pgvector, scoring engine) deployed to AWS ECS via CI/CD
- Own the migration of newsletter ingestion from Optipub to Customer.io in the RAG chatbot service, building an async HTTP client with guarded cursor pagination and typed errors, plus an adapter into the shared content model
- Built a read-only parity check confirming both providers return identical newsletter content before cutover

**Scalean Technologies**: Pune, Maharashtra (Remote) Jun 2026 – Aug 2026

*Backend Engineer Intern*

- Built a GPU-accelerated OCR-KYC pipeline with PaddleOCR (CUDA) and FastAPI, cutting extraction to **1–2s per document** via in-memory model loading after benchmarking 3 engines (Tesseract, EasyOCR, PaddleOCR)
- Extracted structured KYC fields (name, DOB, address) from ID documents using regex and bounding-box line grouping, behind async file uploads, strict input validation, and SMS + email OTP verification with brute-force protection
- Architected a microservices backend from scratch, an **API Gateway + 4 services** with Consul service discovery, JWT auth, SlowAPI rate limiting, CORS, and API versioning, plus a React + TypeScript 5-step onboarding frontend

**XeTech**: Noida, Uttar Pradesh Jun 2025 – Jul 2025

*Web Developer Intern*

- Built responsive React.js UI components (JSX, props, state) for desktop and mobile, shipped via Git and GitHub review

## Projects

**SmartRoute**: Intelligent LLM Gateway github.com/AnmolSrivastavaaaa/SmartRoute

*FastAPI, PostgreSQL, Redis, React, TypeScript, Groq, Gemini*

- Architected a full-stack LLM routing gateway that classifies request complexity and routes across 2 providers and 3 cost tiers, **cutting inference spend 58%**, with output quality verified by an independent cross-provider benchmark
- Built cross-provider failover with a circuit breaker that opens after 3 consecutive failures (60s cooldown), validated by a live chaos test that revoked a real provider API credential mid-run and confirmed fully automatic recovery
- Implemented tenant-isolated Redis caching that **cut repeat-query latency 97% (1555ms to 42ms)** and that request's cost to $0, plus per-key Redis fixed-window rate limiting verified to trip at exactly the 30th request per minute
- Wrote **18 automated tests (100% passing)** covering routing, quality validation, and failover, including mocked chaos tests, plus a JWT-secured React + TypeScript dashboard tracking cost savings, cache hit rate, and p50/p95/p99 latency

**SmartDocs**: Private Document Q&A (RAG) github.com/AnmolSrivastavaaaa/SmartDocs

*FastAPI, ChromaDB, Sentence-Transformers, Hugging Face, PyTorch*

- Built a private document Q&A app on RAG: Sentence-Transformers embeddings, ChromaDB search, grounded answers
- Eliminated external LLM API cost and data exposure by running GPT-2 locally (Hugging Face, PyTorch) behind FastAPI

## Leadership

**Music Band, United University**: Prayagraj, Uttar Pradesh

*Founder & Band Leader*

- Founded and led a college band end to end, recruiting members, planning rehearsals, and directing live performances

## Skills and Interests

- **Backend:** Python, FastAPI, Async Python, REST APIs, Microservices, JWT, SQLAlchemy, PostgreSQL, Redis
- **AI / ML:** LLMs, RAG, Embeddings, ChromaDB, pgvector, Hugging Face, PyTorch, PaddleOCR, Groq, Gemini
- **Frontend & DevOps:** React, TypeScript, JavaScript, HTML/CSS, Java, R, Git, CI/CD (Bitbucket), AWS ECS, Sentry
- **Interests:** Guitar, flute and piano (2 years of classical music training); sketching; swimming